TutorAdvice

Summary

TutorAdvice is a tool designed for students considering postgraduate studies, helping them evaluate their eligibility based on academic performance while offering tutoring services to improve grades. Overall, it calculates grade averages, simulates potential improvements, creates tailored study plans, and helps manage tutoring costs efficiently.

The program begins by asking if the user is still studying or has already graduated. If the user is still a student, they can input their grades to calculate a current average. Graduates can directly enter their final average to determine which postgraduate programs they qualify for. For students who are still studying, the program provides the option to simulate potential grade improvements by replacing the three lowest grades with perfect scores, demonstrating how the average could change.

If student opts to improve their grades, they can then use the tutoring service. This involves creating a study schedule by choosing specific days and hours for sessions, with limit of up to 10 hours per week. The program calculates the costs and generates a receipt. Finally, the user can ask to see the receipt and the schedule.

Example use of the project requirements:

- Comparisons and logical operations & If-else statements:
  The code uses comparisons and logical operations, as well as if-else statements to give an appropriate response to chosen options and to check if the user's current average meets the minimum grade requirements for schools of various levels. For example, if the average is less then 5.5, the user is told that their options are limited.

- Lists and at least 3 different list manipulations
  To store the manually entered grades and then perform calculations for improved averages and tutor services' price, a list "list_grades" is used. The manipulations performed are the following:
    1. list_grades.append: add user inputs to the list of grades

2. list_grades.extend: add multiple prerequired grades to the list
3. list_grades.remove: find the minimum grades eliminating them from the list during the process

- For and while loops & try statements:
  While loops and try statements are frequently used to manage user input and ensure proper data entry. For example, the program repeatedly asks the user to enter grades until a minimum of three grades is collected. For loops are used to iterate through items in a list as a part of the functions, described below.

- Input(), print(), and typecasting:
  The input function captures user responses for various prompts, such as entering grades, selecting options, or specifying study hours. The data is typecasted into the required format, if needed (for instance, floats are needed for most calculations). The print function is used to display calculated averages, eligibility results, costs, etc.

- Function implementations:
  Since there are repetitive calculations, the functions are used. For instance, there is a function used for calculating the user's average. Additionally, the eligibility check, using comparisons, is also similar throughout different steps of the project, therefore, a function is used for this as well.

- Write and read from files:
  The program uses commands from the io module to record study schedules and tutoring costs, as well as providing the user the option to view their study plan, once the schedule is completed.

Authorship declaration

I hereby declare that the work I am submitting is entirely my own. I have not used any generative AI tools for any part of the code or text in my project. Additionally, I did not rely on any external sources or existing code, and therefore, I am not providing acknowledgments or references. My work adheres to the university's code of conduct.